$$\left.\begin{aligned} v_a' &= \frac{v_0}{2}\cos\omega_1 t \\ v_b' &= -\frac{v_0}{2}\cos\omega_1 t \\ v_c' &= \frac{v_0}{2}\cos\omega_2 t \\ v_d' &= -\frac{v_0}{2}\cos\omega_2 t \end{aligned}\right\} \qquad (2)$$

$$\left.\begin{aligned} X_a' &= -\frac{l}{2} + \frac{v_0}{2\omega_1}\sin\omega_1 t \\ X_b' &= \frac{l}{2} - \frac{v_0}{2\omega_1}\sin\omega_1 t \\ X_c' &= \frac{l}{2} + \frac{v_0}{2\omega_2}\sin\omega_2 t \\ X_d' &= \frac{3l}{2} - \frac{v_0}{2\omega_2}\sin\omega_2 t \end{aligned}\right\} \qquad (3)$$

$$\left.\begin{aligned} X_a &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_1}\sin\omega_1 t - \frac{l}{2} \\ X_b &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_1}\sin\omega_1 t + \frac{l}{2} \\ X_c &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_2}\sin\omega_2 t + \frac{l}{2} \\ X_d &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_2}\sin\omega_2 t + \frac{3l}{2} \end{aligned}\right\} \qquad (4)$$

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_1 t \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_1 t \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_2 t \\ v_d &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_2 t \end{aligned}\right\} \qquad (5)$$

$$\omega_1 = \omega_2 = \sqrt{\frac{2k}{m}} = \omega_0 \qquad (6)$$